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A new attention-based CNN approach for crop mapping using time series Sentinel-2 images

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ABSTRACT

Accurate crop mapping is of great importance for agricultural applications, and deep learning methods have been applied on multi-temporal remotely sensed images to classify crops. However, due to the geographic heterogeneity, the spectral profiles of the same crop can vary spatially, and thus using the spectral features alone can limit the model performance in mapping crops in large scales. Moreover, it is a challenge for traditional deep learning models to accurately capture the important information from a large number of features. To address these issues, in this study, we developed a novel attention-based convolutional neural network (CNN) approach (Geo-CBAM-CNN) for crop classification using time series Sentinel-2 images. Specifically, geographic information of crops was first integrated into an advanced attention module, Convolutional Block Attention Module (CBAM) to form a Geo-CBAM module which can help mitigate the impacts of geographic heterogeneity and restrain unnecessary information. Then, the developed Geo-CBAM module was embedded into a CNN model to boost the model's attention both spectrally and spatially. The proposed Geo-CBAM-CNN model was validated on four main crops over six counties with different geographic environments in the U.S. Also, it was compared to three other state-of-the-art machine learning approaches, including CBAM-CNN, CNN and Random Forest (RF). The results showed that the proposed model achieved the best performance, reaching 97.82% overall accuracy, 96.82% Kappa coefficient and 96.96% Macro-average F1 score. Moreover, the developed Geo-CBAM-CNN model showed strong spatial adaptability, indicating its superior performance in large scale applications. Furthermore, by visualizing the structure of the Geo-CBAM-CNN, we found that the model automatically allocated different weights to the features, and generally, the red-edge features in the middle of the year obtained more attention.

1. Introduction

The growing population significantly increases the pressure on the food system (Bajželj, 2014; Tilman et al., 2011), leading to the intensification of agricultural productions (Bargiel, 2017). Agricultural intensification is typically accompanied by increased use of fertilizers and irrigation, and thus poses a threat to ecological processes (Matson et al., 1997; Foley et al., 2005). Therefore, it is essential to optimize agricultural planning and management in order to avoid land degradation and water pollution. For this purpose, detailed spatial information on agricultural land use is required for planners and decision-makers. Due to the obvious interannual variation in crop types on agricultural land, accurate crop mapping remains an indispensable and challenging task.

Traditionally, crop type information was mainly obtained through field surveys (USDA, 2020) which are laborious and expensive, especially when mapping crops in large areas (Carletto et al., 2015; Gourlay et al., 2017). The increasing availability of remote sensing data has provided opportunities to create crop maps automatically based on spectral differences between crops (Waldner et al., 2015). In this context, some researchers employed single-date remote sensing data to identify crops, and the imagery date was selected using the crop calendar. For example, an image acquired on September 22 from the Indian Remote Sensing Satellite (IRS-1D) was adopted to identify the spatial distribution of cotton and rice in Punjab, India (Mathur and Foody, Apr.

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2008). Also, a SPOT5 multispectral image acquired on 16 May 2006 was used to classify crops in south Texas (Yang et al., Feb. 2011). Although these approaches require fewer image processing operations, determining the optimum crop discrimination period is rather challenging, especially for the crops with similar spectral signatures. To improve the classification accuracy, times-series images have been applied to characterize the crop growing dynamics (Hao et al., 2020). For example, Landsat imagery was used to classify agricultural landscapes in Egypt, and multi-temporal data was demonstrated to be more effective than single date data (Pax-Lenney, 1997). Compared with Landsat satellite data, Sentinel-2 data produced a significant improvement in land cover classification accuracy, which could be attributed to the finer spatial, spectral and temporal resolutions (Forkuor et al., 2018). Multi-temporal data from Sentinel-2 were used to classify land-use in south-central Sweden, and the results indicated that the red-edge bands dominated by scenes from spring and summer were important for the classification accuracy (Abdi, Jan. 2020). Also, the use of dense time-series imagery provided by Sentinel-2 alleviated the issue of identifying the optimal crop discrimination period, and improved the classification accuracies of nine crops in Austria compared with using single-date data (Vuolo et al., 2018).

Instead of using remote sensing data directly, some studies extracted phenology features from the time-series images. For example, twelve satellite-derived metrics including the time of peak normalized difference vegetation index (NDVI) and maximum NDVI were successfully used for interannual phenological variability measurement in the U.S. (Reed et al., Oct. 1994). Another study used the land surface temperature (LST) to extract the transplanting phase of rice automatically, and then collected the NDVI, enhanced vegetation index (EVI) and land surface water index (LSWI) in the phase to establish an algorithm for identifying the rice in northeastern Asia (Dong, 2016). More complex function-based temporal feature extractors have also been used in crop classification and phenology studies (Olsson, 1994; Geerken, 2009; Zhong et al., 2019; Bradley and Jacob 2007; Badhwar, Jan. 1984). For example, double asymmetric sigmoid functions were applied to fit VI profiles, and the final parameters of the functions representing the phenological metrics were then used as model inputs for crop classification in California (Zhong et al., 2011). Despite successful, most existing feature extraction approaches have a limitation in automation (Zhong et al., 2019).

Recently, deep learning has been considered as a breakthrough technology and successfully applied in image classification, image segmentation and natural language processing (Krizhevsky et al., 2012; Badrinarayanan et al. 2017, Young et al. (2018); Hinton, 2012). Compared with the traditional classification methods, deep learning models have deeper and complex structures, which could enable them to automatically learn useful information from a large number of features with multiple levels of representation to achieve good classification accuracy (Schmidhuber, 2015; Lecun et al., 2015). Some scholars have applied deep learning methods in crop classification. For example, long short-term memory (LSTM) performed satisfactorily in a long time series land cover classification in China, achieving an 84.2% overall accuracy (Wang et al., Jul. 2019). Another study showed that the convolutional neural network (CNN) outperformed multilayer perception (MLP) and Random Forest (RF), with an accuracy of more than 85% for all crops in Ukraine (Kussul et al., 2017). In addition, the CNN was compared with LSTM and three other traditional machine learning algorithms for crop classification in California, and the results indicated that CNN performed best with an 85.54% accuracy (Zhong et al., 2019).

To further optimize the deep learning models, the attention mechanism has been introduced to help reduce the large amounts of intermediate features extracted from the hidden layers, by addressing more attention to the informative features. The attention mechanism is inspired by the cognitive function of human beings, which selectively focuses on prominent parts to capture the data structure better, instead of processing a whole image at once (Itti and Koch, 2001). Most recently, some studies have incorporated the attention mechanism to enhance the performance of CNN in classification tasks. For instance, using attention mechanism and residual network structure, a Residual Attention Network was designed to generate attention-aware features for better classification (Wang, et al., 2017). Rather than focusing on feature attention, a Squeeze-and-Excitation network (SEnet) was proposed by using global average pooling to squeeze the intermediate features along the channel dimension to capture channel-wise attention. Based on the SEnet, a Convolutional Block Attention Module (CBAM) was designed considering both spatial and channel-wise attention. The results showed that the proposed CBAM focused on the target properly and produced finer attention than SEnet (Woo et al., 2018). In the field of crop classification, the attention mechanism can help detect the phenological differences of crops from temporal-spectral features, and thus improve the classification accuracy. For example, SEnet was adopted to process multi-temporal spectral features and improved the crop classification accuracy in a California valley (Li et al., 2020).

Due to different precipitation, climate conditions and soil types across large areas, the phenology of the same crop can vary over space (Hou, 2014; Wang et al., 2016). For example, Liu et al. conducted experiments from 2007 to 2011 at 34 sites in China to examine differences in phenological responses of maize to latitude, the results demonstrated a significant and positive relationship between both latitude and photoperiods and latitude and total leaf number (Liu, 2013). Because spectral profiles are the spectral responses to crop phenology, profiles of the same crop may differ in different areas. Using only spectral profiles, classifiers may classify the same crop from different regions as different types. Therefore, the geographic information is vital for accurate crop classification. However, previous studies mainly focused on using the spectral features for crop classification, without considering the geographic heterogeneity which is crucially important for large-scale crop mapping. To address this issue, in this study, we developed a spatiotemporal attention-based CNN approach (Geo-CBAM-CNN) to map crops using multi-temporal Sentinel-2 data. Specifically, a Geo-CBAM module was developed by incorporating the geographic information into the CBAM to address attention spatially. Then, the Geo-CBAM-CNN model was designed by combining the Geo-CBAM module with the residual blocks (ResBlocks) (He et al., 2016) which could help gain accuracy from greatly increased depths. Finally, the developed approach was compared with CNN and RF, and the important features were visually interpreted by the gradient-weighted class activation mapping (Grad-CAM) tool (Selvaraju et al., 2017).

2. Materials and methods

2.1. Study area

As a leading agriculture country in the world, the U.S. produces a wide variety of agricultural products across the nation. This study focused on four major crops: corn, cotton, soybean and winter wheat, which account for over 70% of the planted acreage in the U.S. in 2018 (Nass, 2018), and their primary production regions include the State of Oregon, Midwestern States and States along the Mississippi River (Fig. 1). To validate our approach in crop mapping, we selected six counties in the production regions of the four main crops, including Sherman (Oregon), Traill (North Dakota), Decatur (Kansas), Haskell (Haskell), Randolph (Indiana) and Mississippi (Arkansas). In addition, the crop planting and harvesting progress at each location is shown in Fig. 2, and differences were observed across species and regions. By covering a wide range of growing conditions, the six counties thus provide an opportunity to assess the model large-scale applicability.

2.2. Data acquisition and preprocessing

2.2.1. Ground reference

The Cropland Data Layer (CDL), produced by the USDA NASS, is an

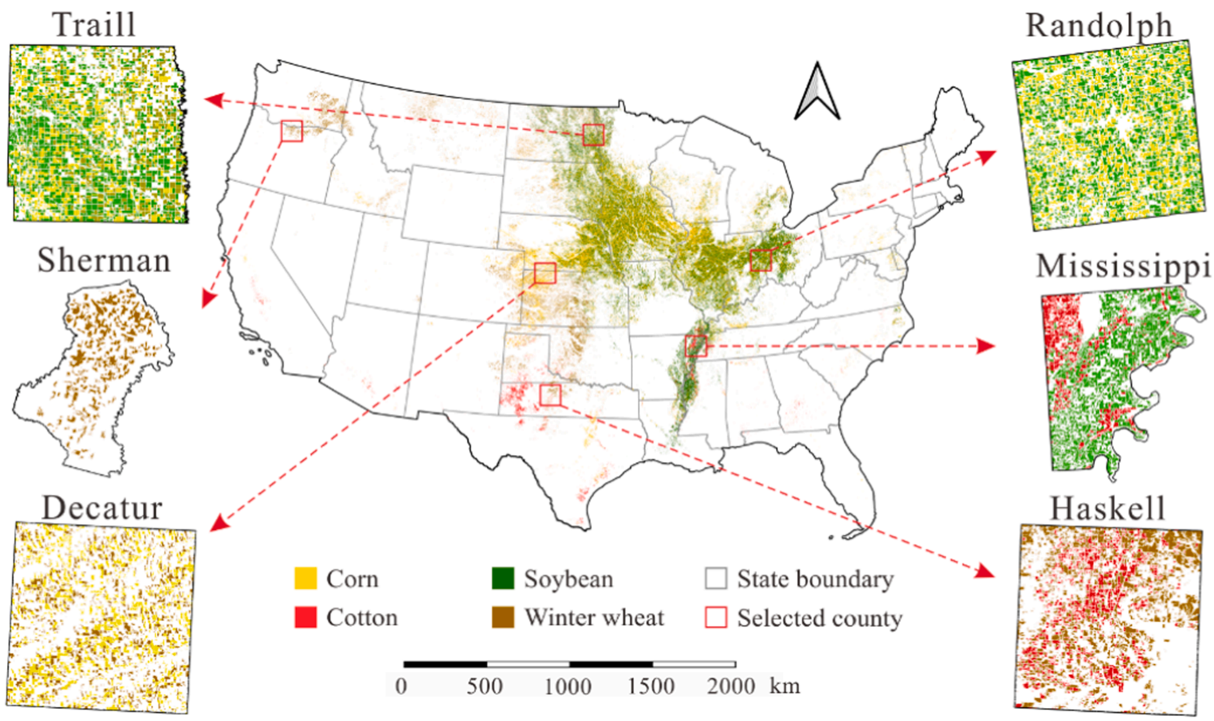

Fig. 1. Corn, cotton, soybean and winter wheat growing area in the U.S. in 2018.

annual land cover map with a 30 m spatial resolution (Boryan et al., 2011). Derived from Landsat images, the CDL consists of more than 100 land cover types, and provides classification confidence for each pixel. In this study, we constructed the ground reference data using the CDL in 2018 for model development. To select high-confidence data as the ground truth, we used 90% as the confidence threshold to mask the CDL layer. Due to the high-resolution of the CDL, our research area in total had around 10 million pixels, leading to a huge computational challenge. To reduce the computational load, we downscaled the CDL map to a 90 m resolution by aggregating the neighboring pixels using a 3x3 window. The new samples with 90 m resolution were labeled as specific crops if the nine pixels within the window are homogenous. Otherwise, they were not considered. Since this study focused on corn, cotton, soybean and winter wheat, the other crop types were labeled as the "other" class.

2.2.2. Sentinel-2 data

Sentinel-2 satellite images were used to extract spectral information of the crops in the study area. The full sentinel-2 mission comprises two satellites (S2A and S2B) which could provide wide-swath and high-resolution multispectral images. A total of 1727 images in the whole 2018 were preprocessed to remove cloud contamination using their cloud mask bands, and ten spectral bands of Sentinel-2 images ranging from the visible, near-infrared (NIR) to the short wave infrared (SWIR) were considered in this study. The details of each band are shown in Table 1.

Specifically, for each type of crop, the ten Sentinel-2 spectral bands were collected at a 10-day interval in the entire 2018, resulting in a total of 36 temporal data records. The 36 time-series features were used in this study, due to the fact that the growing season of the four crops covered almost the entire year. All the spectral bands were spatially aggregated to 90 m which is the spatial resolution of the constructed ground reference map. Due to the cloud contamination, there may be different gaps in time series data, leading to missing features in the data samples. Therefore, samples lacking more than 20% of the time series features are removed to reduce data noise (Bijlsma, 2006). We then filled the missing attributes of the remaining samples using the Savitzky–Golay algorithm (Jönsson and Eklundh, 2002), and the final number of crop samples in each county is shown in Table 2. The data were processed through the Google Earth Engine (GEE) (Gorelick et al., 2017) platform.

2.3. Methodology

2.3.1. Convolutional Block attention module (CBAM)

CBAM, proposed by Woo et al. (Woo et al., 2018), is an effective attention module that can be embedded in any CNN to boost its representation power. CBAM aims to guide the CNN model to focus on informative features and restrain unnecessary information, and its architecture is shown in Fig. 3. Since the convolution operation extracts features by blending channel and spatial information, CBAM is designed to emphasize the importance of intermediate features (features in the convolution layers of the CNN) along the channel and spatial dimensions by sequentially using two sub-modules: channel attention module and spatial attention module. The architectures of the two sub-modules are respectively demonstrated in Fig. 3(b) and 3(c), and the details of each sub-module are described below.

Channel attention module: Each channel of intermediate features in the convolution process is considered as a feature detector, and the channel attention module is used to exploit the inter-channel relationship to decide which channel is more meaningful. First, the spatial dimension of the intermediate features is squeezed by average-pooling and max-pooling to gather two different context features respectively. Then, the two types of features are transformed by a shared multi-layer perceptron (MLP) and merged using element-wise summation. Finally, the fused feature is activated by a sigmoid function to represent the channel importance of the original feature map. The calculation process of the channel attention modules is shown as:

$$\mathbf{M_c}(\mathbf{F}) = f_{\mathrm{sigmoid}}(\mathrm{MLP}(\mathrm{AvgPool}(\mathbf{F})) + \mathrm{MLP}(\mathrm{MaxPool}(\mathbf{F}))) \tag{1}$$

Spatial attention module: Unlike the channel attention module, the spatial attention module is focused on "where" is an information part that complements the channel attention. It aims to generate refined features to exploit the inter-spatial relation of the original features. First,

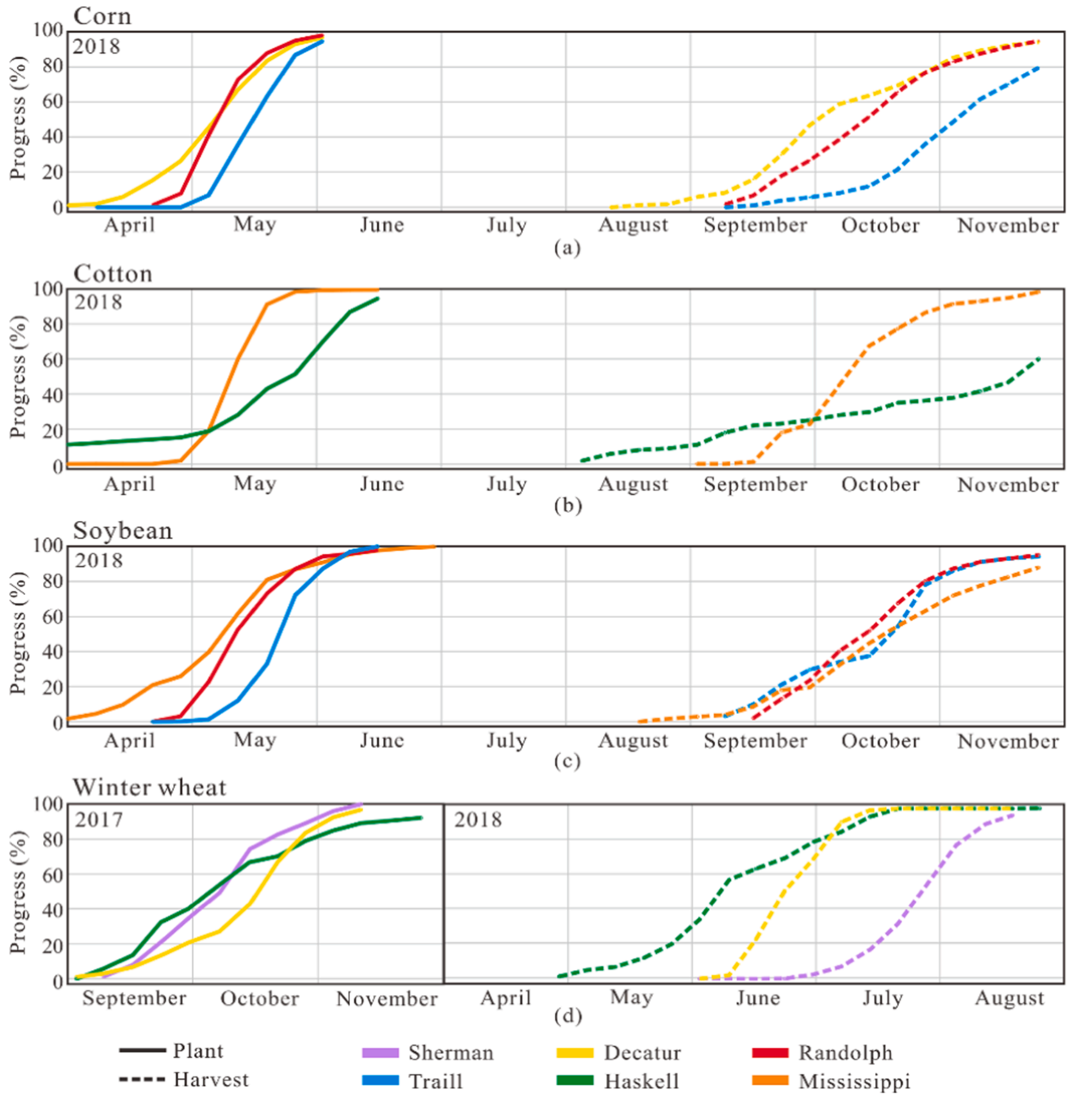

Fig. 2. Crop planting and harvest progress (%) at each location.

Table 1
Ten spectral bands of the Sentinel-2 sensor.

Band #	Name	Spatial Resolution	Wavelength
B2	Blue	10 m	490 nm
B3	Green	10 m	560 nm
B4	Red	10 m	665 nm
B5	Red Edge 1	20 m	705 nm
B6	Red Edge 2	20 m	740 nm
B7	Red Edge 3	20 m	783 nm
B8	NIR	10 m	842 nm
B8A	Red Edge 4	20 m	865 nm
B11	SWIR 1	20 m	1610 nm
B12	SWIR 2	20 m	2190 nm

Table 2
The number of crop samples in each county in 2018.

	County name	# Samples in each county
Corn	Decatur	9097
	Randolph	16,558
	Traill	38,624
Cotton	Haskell	29,162
	Mississippi	7191
Soybean	Randolph	19,521
	Mississippi	4562
	Traill	73,281
Winter wheat	Decatur	7644
	Haskell	60,982
	Sherman	42,329
Other	Decatur	22,943
	Haskell	34,854
	Mississippi	10,308
	Randolph	3229
	Sherman	159,708
	Traill	67,310

average-pooling and max-pooling are applied along the channel axis of the original features. The outputs of the two operations are concatenated to produce a new feature descriptor. Then, the concatenated descriptor is converted to generate a spatial attention map through a 7×7 convolution and a sigmoid operation. This process can be computed as:

$$\boldsymbol{M}_s(\boldsymbol{F}) = f_{sigmoid}(Conv([AvgPool(\boldsymbol{F}); MaxPool(\boldsymbol{F})])) \quad (2)$$

Finally, CBAM infers attention maps by combining the channel and spatial attention modules, and then generates refined feature maps. The combination of the two sub-modules are described as:

$$\boldsymbol{F}' = \boldsymbol{M}_c(\boldsymbol{F}) \otimes \boldsymbol{F},$$

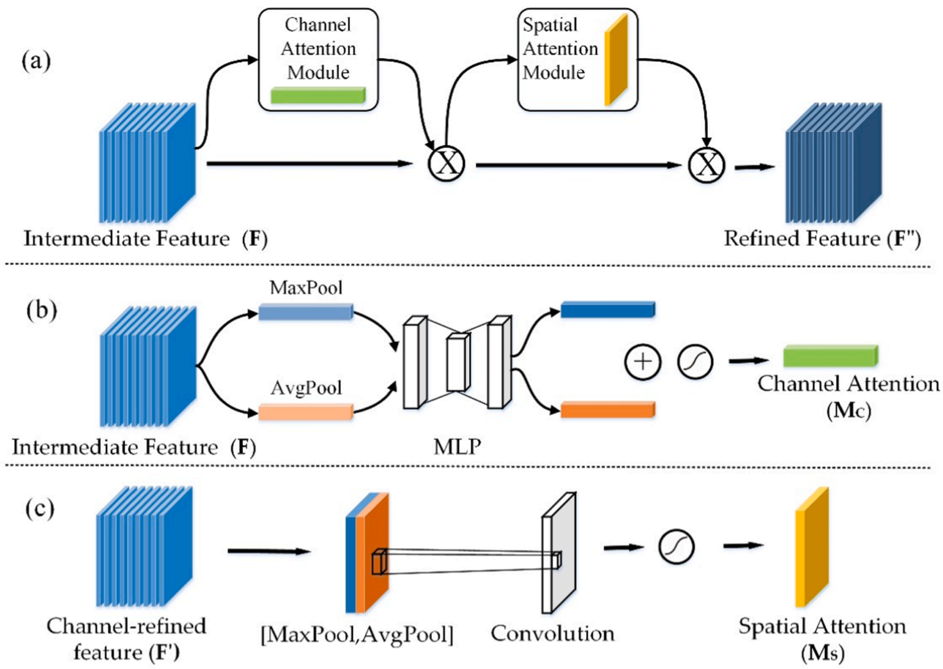

Fig. 3. Architectures of CBAM and its sub-modules: (a) CBAM; (b) channel attention module; (c) spatial attention module. F represents an intermediate feature map of CNN; F' is the refined feature map before spatial attention module; F'' is the refined feature map after CBAM; Mc represents the extracted features after channel attention module, and Ms is the extracted features after spatial attention module.

$$\boldsymbol{F}'' = \boldsymbol{M}_s(\boldsymbol{F}') \otimes \boldsymbol{F}' \quad (3)$$

where **F** represents an intermediate feature map of CNN; $\otimes$ denotes element-wise multiplication; and **F**'' is the refined feature map.

2.3.2. Geo-CBAM based CNN

As an attention model, CBAM can help strengthen CNN models by enhancing the differences among the spectra of different crops, leading to improved classification performance. However, as mentioned in the introduction, spectral patterns can vary spatially even for the same type of crop due to different climate conditions, soil types and management practices. The complex spectral patterns may affect model accuracy. To address the geographical heterogeneity, a new attention model Geo-CBAM is developed in this study by incorporating the geographic information into the CBAM. Specifically, the center point geographic coordinates (latitude and longitude) of each crop sample were extracted and embedded into the channel attention sub-module of the CBAM, and the designed architecture is shown in Fig. 4. There are two reasons for incorporating geographic information into the channel attention module. First, in this study, since the total number of input features is quite large (360 input features in total), adding the two-dimensional geographic coordinate features (latitude and longitude) directly into the input features would not make much difference. Second, the intermediate feature (F) in the channel attention module are high-level features extracted from the convolutional layers. After average pooling and max pooling, those features will be further reduced and enable finer attention inference (Woo et al., 2018). By adding geographic information to the channel module, the geographic information is combined with the reduced features after the pooling operations (MaxPool and AvePool) and thus can exhibit a more important role than directly adding to the original high dimensional features. The following MLP module can help further address the attention inference. Besides, since the channel and spatial attention sub-modules are sequentially connected, the influence caused by the modified channel attention can also be transferred to the subsequent spatial attention, and thus benefits the whole module with addressing attention both spectrally and spatially.

For the classification task, a Geo-CBAM-CNN model was then developed by combing the Geo-CBAM module and ResBlocks. The architecture of the model is demonstrated in Fig. 5, and the input time-series spectral features were converted to a 2D image to account for the spectral-temporal dynamics. The model consisted of eight hidden layers, including a normal convolution, three ResBlocks, three Geo-CBAMs and a global average pooling layer (GAP). ResBlock is the base unit of ResNets, which could reduce the optimization issues of deep networks and showed remarkable performance (He et al., 2016). In this study, the ResBlock contained five stacked layers and a convolution shortcut connection which could help avoid gradient diffusion in deep learning models (He et al., 2016). Batch normalization layers were

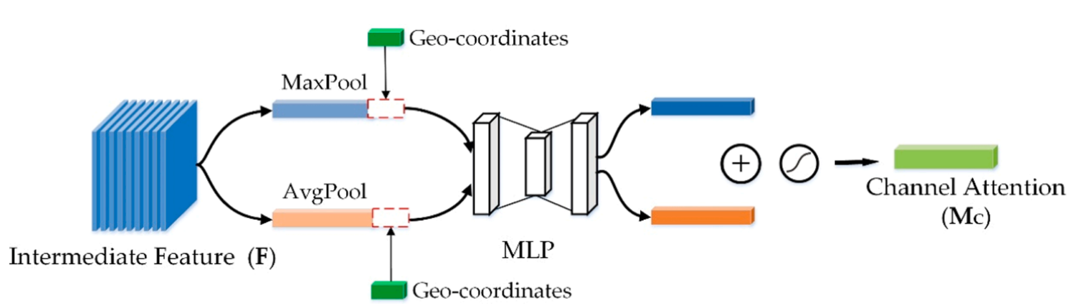

Fig. 4. The architecture of the designed channel attention sub-module in the Geo-CBAM.

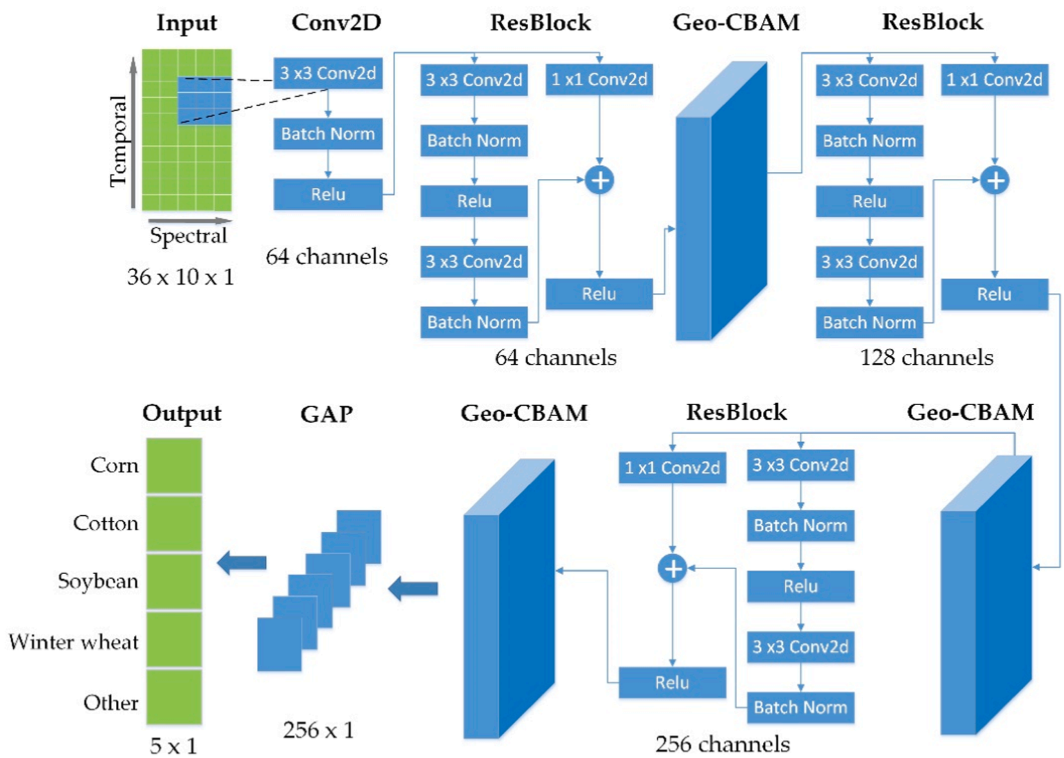

Fig. 5. The architecture of the developed Geo-CBAM-CNN model.

applied in the model, because it could reduce the internal covariate transition of the neural network to accelerate model convergence (Ioffe and Szegedy, 2015). The output layer was a dense layer with a softmax activation which learned the probability distributions over the five classes. Finally, the type of the crop sample was determined by the highest probability class. The batch size, convolution activation function and optimizer were set to 128, Rectified Linear Unit (ReLU) (Nair and Hinton, xxxx) and Adam (Kingma and Ba, 2015) in our model.

2.3.3. *Comparison approaches and evaluation metrics*

To evaluate the model robustness with a small training set, 99% of all the data samples were randomly selected for model testing, and the rest 1% samples were further divided into two parts: 70% used for training and 30% for validation. Therefore, 0.7% of all the available data are used for training, 0.3% are for validation, and 99% for testing. The proposed method Geo-CBAM-CNN was compared to three other approaches, including CBAM-CNN, CNN and RF. For a fair comparison, the two CNN based deep learning models were designed to have a similar structure with Geo-CBAM-CNN, while the CBAM-CNN used the CBAM attention mechanism and CNN had no attention module. In addition, RF, which has been applied in various studies, was also included as a baseline approach in this study. It was constructed by multiple decision trees and the final prediction was determined by combining the results from all the individual trees. The three deep learning models were implemented in Tensorflow 2.0 (Abadi, et al., 2016), and the RF was developed using the Scikit-learn package in Python (F. Pedregosa FABIANPEDREGOSA et al., "Scikit-learn: Machine Learning in Python Gaël Varoquaux Bertrand Thirion Vincent Dubourg Alexandre Passos PEDREGOSA, VAROQUAUX, GRAMFORT ET AL. Matthieu Perrot,", et al., 2011). The extracted time-series spectral features were used to develop the four models, and the Geo-CBAM-CNN additionally utilized geographic information. Three metrics were used to assess the classification accuracies, including (i) overall accuracy (OA); (ii) Kappa statistic; (iii) Macro-average F1, the average of F1 scores over all the classes (Zhong et al., 2019). In addition, the McNemar's test was also used to analyze the statistical significance of the results obtained by two models being compared, and the test is calculated as follows. :

$$M = \frac{f_{12} - f_{21}}{\sqrt{f_{12} + f_{21}}} \tag{4}$$

where f_{12} is the number of samples that correctly classified by the proposed method and misclassified by the comparison approach, and$f_{21}$is defined analogously. If$|M| > 1.96$, the differences between the two models are considered statistically significant.

To investigate the attention mechanism in the developed model, we used the Grad-CAM to visually explain the Geo-CBAM-CNN. Specifically, the Grad-CAM was used to produce a localization map that highlights the important features for classifying a specific crop, and this was achieved by applying the Grad-CAM on the last convolutional layer for assigning the important values to each neuron for a particular class.

3. Results

3.1. *Spectral profiles of crops*

For the four main crops, their time-series spectral profiles are shown in Fig. 6, with each color representing an individual county. The figure showed that for the same crop, the profiles varied from county to county and band to band. For example, the cotton in Haskell and Mississippi showed a clear difference in the two SWIR bands (Fig. 6 (b9)-(b10)), while large overlaps were observed in the Red Edge and NIR regions (Fig. 6 (b5)-(b8)), indicating their significance in describing the crop-specific characteristics. On the other hand, different crops could present similar patterns in all the bands even in the same county, such as the corn and soybean in Traill or Randolph (Fig. 6a and 6c), introducing challenges in distinguishing the two crops. Therefore, to develop a robust classifier for large-scale crop classification, it is necessary to address attention in both the spectral and spatial domains.

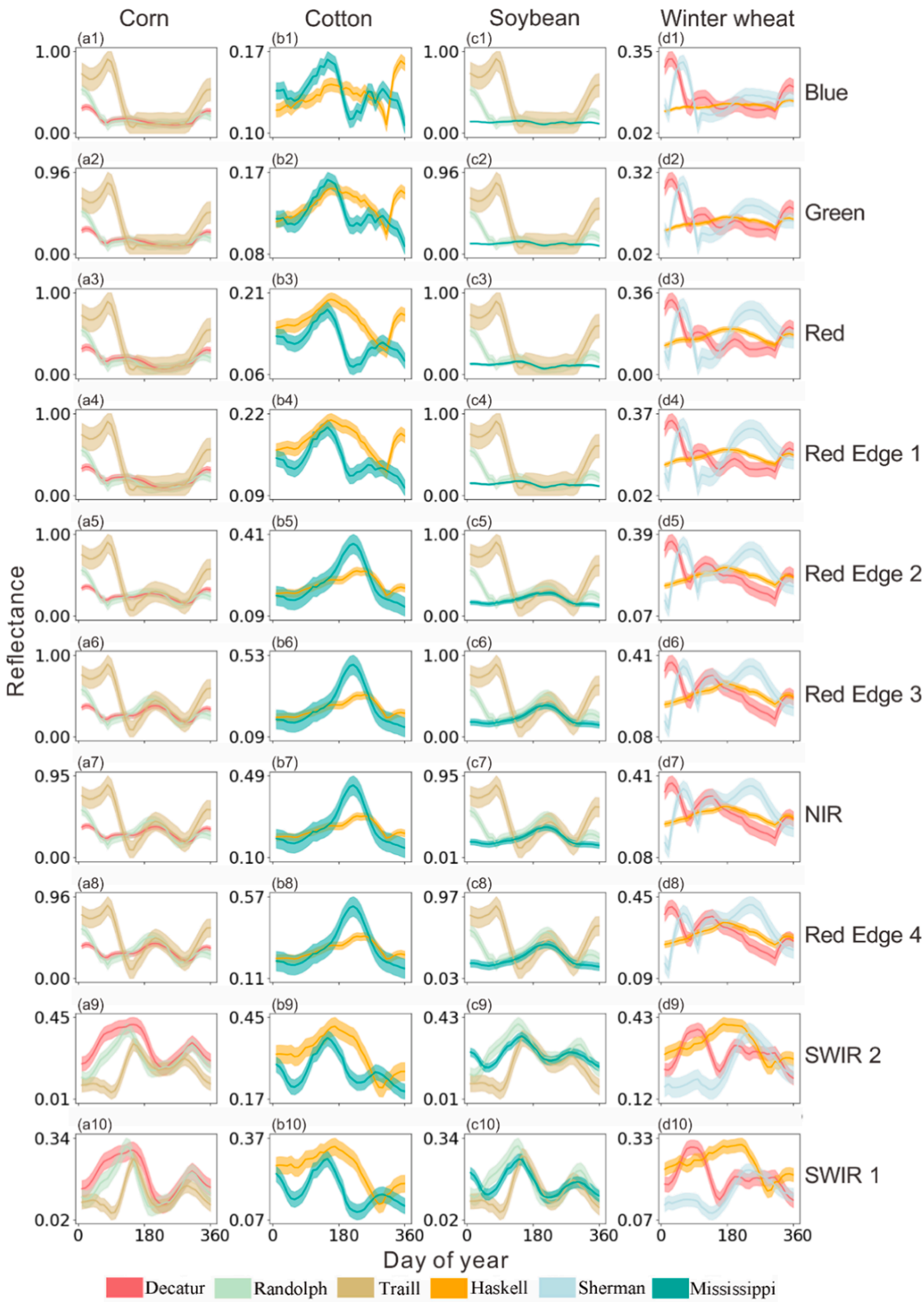

Fig. 6. Time-series spectral profiles in different counties from (a) Corn; (b) Cotton; (c) Soybean; (d) Winter wheat with 10 spectral bands (1–10): Blue, Green, Red, Red Edge 1, Red Edge 2, Red Edge 3, NIR, Red Edge 4, SWIR 1, SWIR 2. The lines represent the average values of reflectance, and the buffers refer to one standard deviation from the average values. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2. Model performance comparison

The classification accuracies of different models are shown in Table 3. Among all the approaches, the Geo-CBAM-CNN obtained the best performance, achieving 97.82% OA, 96.82% Kappa and 96.96% Macro-average F1. The CBAM-CNN was the second best method, demonstrating the efficacy of incorporating the attention mechanism into CNN. As the only non-deep-learning model, the RF achieved lower accuracies than others. Besides, we used McNemar's test to analyze the statistical significance of the accuracy improvement obtained by the Geo-CBAM-CNN, and the results are shown in Table 4. The calculated statistics are all above 1.96, showing that differences in classification accuracies between the Geo-CBAM-CNN and others are statistically significant.

For further comparison, Fig. 7 demonstrates the accuracy assessments for each class. The four models all achieved satisfactory performances in the "Other" class, with accuracy near 100% for each approach. Specifically, the RF was still inferior to the deep learning models, while there was little variation among the three deep learning methods. This was mainly because the number of samples in this class accounted for nearly 50% of all the training data, providing sufficient samples for training the deep learning models. Regarding the four major crops, the classification accuracies of different models varied clearly. In particular, the proposed method outperformed the other models for all four crops. Cotton exhibited lower accuracies than other crops, and this is likely due to the relatively small quantity of training samples for cotton. Also, it is worth noting that Geo-CBAM-CNN and CBAM-CNN achieved much higher accuracies than CNN and RF in cotton, indicating the effectiveness of using attention mechanism with a relatively small training set.

3.3. Model spatial adaptability

We further reported crop classification accuracies in different counties in Table 5, and the highest accuracy was highlighted in bold. The results showed that the proposed Geo-CBAM-CNN model exhibited stable performance across counties for each crop, achieving more than 90% accuracy in all the cases. The CBAM-CNN also showed relatively good performance in most instances (greater than90% accuracy), except for soybean in Mississippi. On the contrary, the CNN and RF performed much worse than the other two attention-based methods. For example, the classification accuracy of the soybean in Mississippi was only 53.57% in RF and 62.76% in CNN. This is likely due to the inconsistent spectral patterns caused by geographic heterogeneity. The strong spatial adaptability shown by the proposed model demonstrates the efficacy of embedding the spatial information into the attention mechanism. To further compare the classification performances of different models in the spatial domain, we chose Haskell county as an example and showed the classification error map for each model in Fig. 8. It showed that much fewer areas were misclassified in the proposed Geo-CBAM-CNN model (Fig. 8 (a)), and the errors were also less concentrated (blue rectangles) when compared to other approaches.

3.4. Feature importance visualization

The Grad-CAM visualization results are shown in Fig. 9, with warmer color representing a higher importance value.

Strong features can be easily identified from Fig. 9 for each crop. In the temporal domain, the features in the crop growing season obtained more attention, especially those near the mature period. Also, we noticed that the important time window located within 90–180 DOY (day of the year) for winter wheat, which was earlier than other crops. This time-shifting is mainly because unlike other crops which typically are planted in the late spring, the winter wheat is seeded in the preceding fall and reaped in the current summer (Fig. 2), resulting in different growing seasons. On the other hand, the important times series features were in the range of 150–250 DOY for both corn and soybean, and this is likely due to the similar phenology of the two crops. Moreover, we found that compared to other crops, the important time period spanned longer in cotton, and this may be because the planting/harvest time of cotton varies greatly in different counties (Fig. 2(b)), and common features are less concentrated within the growing period. In the spectral dimension, the red edge bands which are known to be sensitive to plant physiological status (Adelabu et al., 2014); (Clevers and Gitelson, 2013) , showed significant importance for all the four crops. Besides, blue and green bands were also found to be important for soybean and winter wheat in their mature stages (Fig. 9(c)-(d)), and this may be related to the different chlorophyll, carotenoids and water content of crops (Moghimi et al., 2020; Lichtenthaler, 1987).

Table 3
Classification accuracies of different models.

Metrics	RF	CNN	CBAM-CNN	Geo-CBAM-CNN
OA(%)	95.04	96.43	97.13	**97.82**
Kappa(%)	93.24	94.78	95.82	**96.82**
Macro-average F1(%)	93.18	94.92	96.01	**96.96**

Table 4
McNemar's test between the Geo-CBAM-CNN and other comparison methods.

Method	M
Geo-CBAM-CNN versus RF	101.93
Geo-CBAM-CNN versus CNN	54.91
Geo-CBAM-CNN versus CBAM-CNN	34.92

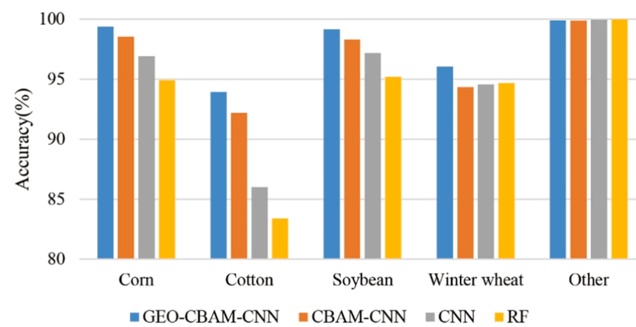

Fig. 7. Crop-specific classification accuracies obtained by different models.

Table 5
Crop-specific classification accuracies in each county obtained by different models.

Crop type	County	RF	CNN	CBAM-CNN	Geo-CBAM-CNN
Corn	Decatur	87.57	90.65	96.95	**98.39**
	Randolph	97.78	96.6	98.71	**99.37**
	Traill	95.29	98.52	98.80	**99.58**
Cotton	Haskell	80.49	84	90.49	**92.61**
	Mississippi	94.38	94.15	98.69	**98.97**
Soybean	Randolph	98.30	98.8	99.11	**99.19**
	Traill	96.86	98.85	99.04	**99.35**
	Mississippi	53.57	62.76	82.84	**95.70**
Winter wheat	Decatur	85.00	85.07	91.24	**94.57**
	Haskell	92.34	92.37	90.81	**93.71**
	Sherman	99.61	99.41	**99.83**	99.55

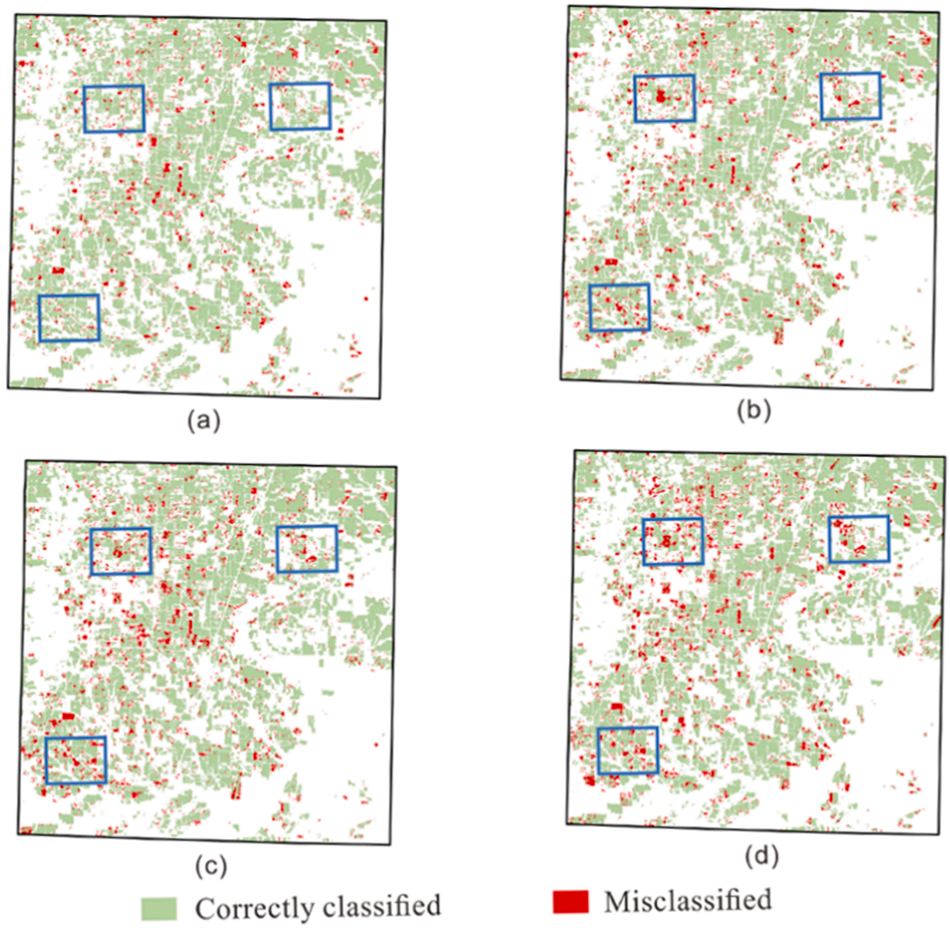

Fig. 8. Spatial distribution of classification errors in Haskell county: (a) Geo-CBAM-CNN; (b) CBAM-CNN; (c) CNN; (d) RF.

4. Discussion

4.1. Performance of the crop classification models

This study developed a new attention-based deep learning model using both spectral and spatial features for large-scale crop mapping. To validate the proposed method Geo-CBAM-CNN, it was compared to three other state-of-the-art machine learning approaches, including CBAM-CNN, CNN and RF. The results showed that the Geo-CBAM-CNN outperformed the other three approaches by more than 1%, 2.04%, and 3.58% in Kappa, respectively. All the improvements were statistically significant, as demonstrated by McNemar‘s test in Table 4. Besides, we also noticed that the CNN based models were superior to the RF model by more 1.5% in Kappa, which was consistent with previous studies demonstrating the efficacy of deep learning in extracting informative features (Zhong et al., 2019); (Lichtenthaler, 1987) . Then, comparing the three CNN methods, better performance was observed in Geo-CBAM-CNN and CBAM-CNN, indicating the advantages of using attention modules in addressing the important features and suppressing the unnecessary information. The best performance exhibited by the Geo-CBAM-CNN model with 97.82% OA, 96.82% Kappa and 96.96% Macro-average F1, further demonstrated the effectiveness of embedding the geographic information into the attention module which can help mitigate the effects of the geographic heterogeneity in large-scale crop mapping.

To estimate the in-season forecasting capability, we developed an experiment to evaluate the classification accuracy of the models against timelines. Especially, a fixed starting date was set at DOY 1, and the ending date was varied from DOY 100 to 360. The reason for this setting is that April (DOY 100) is the mid-season for winter wheat, and the beginning of the growing season for corn, soybean and cotton. For each ending date, the time series features between the starting date and the current ending date were used to train the models. As shown in Fig. 10, the models all performed better with more time-series features until August, and then their accuracies were stabilized. The proposed Geo-CBAM-CNN model performed better than other models, and it achieved 95% Kappa in July, indicating the major crops in the six counties can be accurately classified before harvest.

4.2. Impact of training samples for the classification

To investigate the effects of the number of training data on model performance, we compared the different models by increasing the training sample size from 1% to 10% of the total samples. The Kappa coefficient values obtained by the four models are shown in Fig. 11. It was clear to see that the performance of each model initially increased with more training data, then gradually converged. Compared to the attention-based approaches, the improvement of CNN and RF was more significant, indicating that more training data were needed for these models to become robust. Besides, the proposed model Geo-CBAM-CNN constantly outperformed the other approaches, maintaining a Kappa of greater than 96%. Since collecting the training data can be laborious and expensive, the proposed method had more practical value due to its superior performance with limited training samples.

4.3. Limitations and future work

Although the proposed model achieved good performance in this

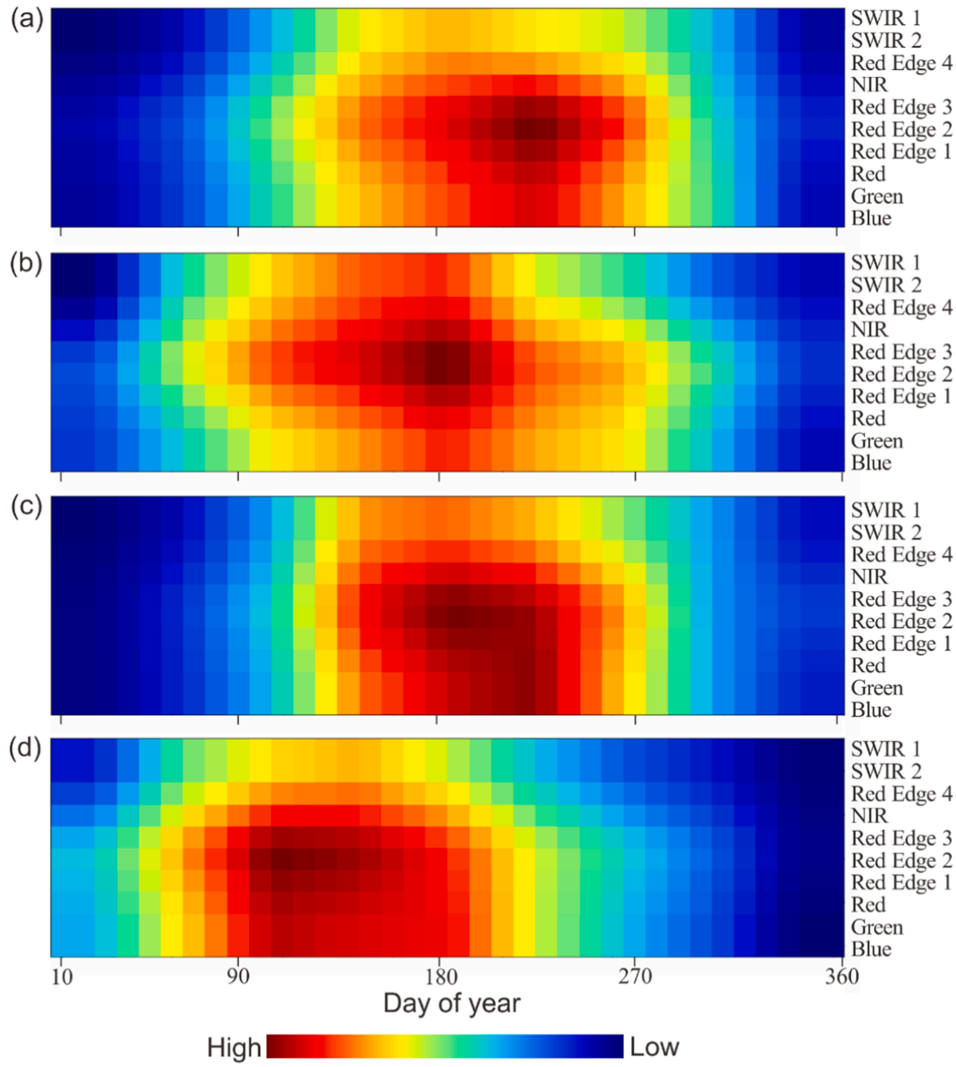

Fig. 9. Crad-CAM visualization results for each crop: (a). corn; (b). cotton; (c). soybean; (d). winter wheat. Red color indicates high weight features, while blue color means low weight features. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

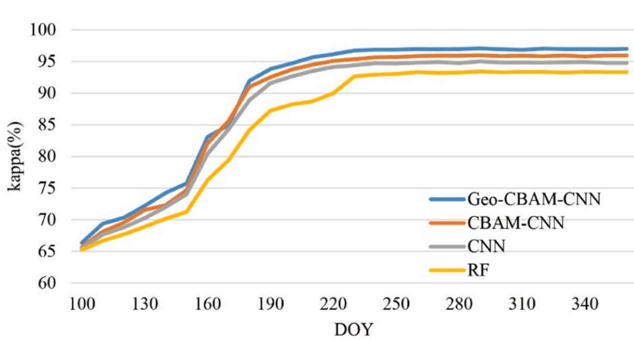

Fig. 10. The performance of crop classification models against timelines.

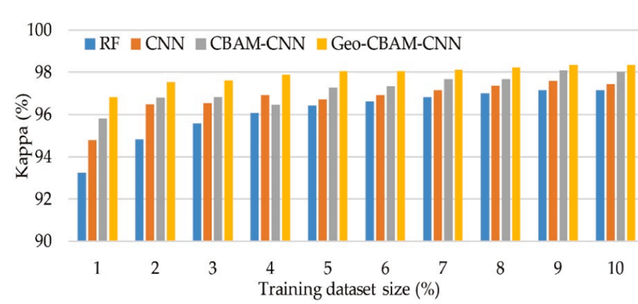

Fig. 11. The performance of crop classification models using different training data.

study, there are still some limitations. First, we only focused on four major crops in the U.S., while more than ten types of crops are grown in some regions, such as California (Li et al., 2020). Future efforts will be devoted to validating the proposed model in more complex regions. Second, to account for the spatial heterogeneity, the simple geographic coordinates were embedded into the model. More advanced approaches, such as geographical clustering, will be investigated to enhance the model generalizability over large areas (Deng et al., Feb. 2018); (Wang et al., 2020) . Also, since the prevalent CNN based models require input data having the same structure, any samples with missing attributes in the temporal domain could not be used in the proposed method. As future work, instead of using only the sentinel-2 data, the harmonized Landsat and Sentinel-2 data with enhanced temporal quality will be explored (Claverie et al., 2018).

5. Conclusion

Accurate crop mapping is of great importance to scientific agricultural planning and management. In our study, a spatiotemporal attention-based CNN model was developed to accurately classify the crop types. Using Sentinel-2 images, we extracted time-series spectral signatures of crops as model inputs. Then, the proposed model was applied to classify four main crops in six counties across diverse regions in the U.S. to test its robustness. We evaluated the model by comparing it with three other approaches, including CBAM-CNN, CNN and RF. The results showed that the proposed model achieved the best in-season performance and showed good spatial adaptability over the six counties, which indicated that this model could generate more reliable crop maps in large areas. In addition, we employed the Grad-CAM tool to examine the focused spectral and temporal information of the model and found that red-edge features in the middle of the year received more attention. Moreover, the proposed model outperformed other models in small sample situations, which is important for scientific and practical uses.

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CRediT authorship contribution statement

Yumiao Wang: Conceptualization, Methodology, Writing - original draft. **Zhou Zhang:** Conceptualization, Resources, Writing - review & editing. **Luwei Feng:** Software, Visualization. **Yuchi Ma:** Data curation, Formal analysis. **Qingyun Du:** Conceptualization, Resources.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

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